Compression Tools

Compression tools are used to compress one or more files or an archive to save space. Once a compressed archive is created, it can be copied to a remote system faster than a non-compressed archive. Compression tools may be used with archive commands, such as *tar*, to create a single compressed archive of hundreds of files and directories. RHEL provides a number of compression tools, such as *bzip2* (*bunzip2*) and *gzip* (*gunzip*), that can be used for this purpose.

Using gzip and gunzip

The *gzip* command creates a compressed file of each of the files specified at the command line and adds the *.gz* extension to each one of them.

To compress files *anaconda-ks.cfg* and *initial-setup-ks.cfg* in the */root* directory, issue the following *gzip* command, and then list them:

```
# gzip /root/anaconda-ks.cfg /root/initial-setup-ks.cfg
# ll /root | grep gz
-rw-------. 1 root root  979 Oct 28 23:04 anaconda-ks.cfg.gz
-rw-r--r--. 1 root root 1011 Oct 28 23:11 initial-setup-ks.cfg.gz
```

To uncompress the files, run either of the following commands on each file:

```
# gunzip /root/anaconda-ks.cfg.gz
# gzip –d /root/initial-setup-ks.cfg.gz
```

Check the files after the decompression using the *ll* command.

Using bzip2 and bunzip2

The *bzip2* command creates a compressed file of each of the files specified at the command line and adds the *.bz2* extension to each one of them.

To compress files *anaconda-ks.cfg* and *initial-setup-ks.cfg* in the */root* directory, issue the following *bzip2* command, and then list them:

```
# bzip2 /root/anaconda-ks.cfg /root/initial-setup-ks.cfg
# ll /root | grep bz2
-rw-------. 1 root root 1086 Oct 28 23:04 anaconda-ks.cfg.bz2
-rw-r--r--. 1 root root 1111 Oct 28 23:11 initial-setup-ks.cfg.bz2
```

To uncompress the files, run either of the following commands on each file:

```
# bunzip2 /root/anaconda-ks.cfg.bz2
# bzip2 –d /root/initial-setup-ks.bz2
```

Check the files after the decompression using the *ll* command.

Archiving Tools

RHEL offers many native tools that can be utilized to archive files for storage or distribution. These tools include *tar* and *star*, both of which have the ability to preserve general file attributes, such as ownership, group membership, and timestamp. The following sub-sections discuss these tools in detail.

Using tar

The *tar* (*tape archive*) command creates, appends, updates, lists, and extracts files to and from a single file, which is called a *tar* file (also a *tarball*). This command has the ability to archive SELinux file contexts as well as any extended attributes set on files. It can also be instructed to compress an archive while it is being created.

tar supports several options, some of which are summarized in Table 2-3.

Switch	Definition
–c	Creates a tarball.
–f	Specifies a tarball name.
–j	Compresses a tarball with bzip2 command.
–r	Appends files to the end of an existing tarball. Does not append to compressed tarballs.
–t	Lists contents of a tarball.
–u	Appends files to the end of an existing tarball if the specified files are newer. Does not append to compressed tarballs.
–v	Verbose mode.
–x	Extracts from a tarball.
–z	Compresses a tarball with gzip command.
--selinux --no-selinux	Includes (excludes) SELinux file contexts in archives.
--xattrs --no-xattrs	Includes (excludes) extended file attributes in archives.

A few examples have been provided below to elucidate the usage of *tar*. Note that the use of the – character is optional.

To create a tarball called */tmp/home.tar* of the entire */home* directory tree:

```
# tar cvf /tmp/home.tar /home
tar: Removing leading `/' from member names
/home/
/home/user1/
/home/user1/.mozilla/
/home/user1/.mozilla/extensions/
```

……. .

To create a tarball called */tmp/files.tar* containing multiple files from the */etc* directory:

```
# tar cvf /tmp/files.tar /etc/host.conf /etc/shadow /etc/passwd /etc/yum.conf
```

To append files located in the */etc/yum.repos.d* directory to *home.tar*:

```
# tar rvf /tmp/home.tar /etc/yum.repos.d
```

To list the contents of *home.tar*:

```
# tar tvf /tmp/home.tar
drwxr-xr-x  root/root       0 2014-10-28  23:04  home/
drwx--x---  user1/user1     0 2014-11-25  10:42  home/user1/
drwxr-xr-x  user1/user1     0 2014-10-29  19:19  home/user1/.mozilla/
. . . . . . . .
drwxrwxr-x  user1/user1     0 2014-11-25  10:42  home/user1/dir1/
drwxr-xr-x  root/root       0 2014-10-29  08:23  etc/yum.repos.d/
-rw-r--r--  root/root      83 2014-10-29  08:04  etc/yum.repos.d/ftp.repo
```

To list the contents of *files.tar*:

```
# tar tvf /tmp/files.tar
-rw-r--r--  root/root       9 2013-06-07  10:31  etc/host.conf
----------  root/root    1177 2014-11-20  10:27  etc/shadow
-rw-r--r--  root/root    1999 2014-11-20  10:27  etc/passwd
-rw-r--r--  root/root     813 2014-04-15  09:54  etc/yum.conf
```

To restore */home* from *home.tar*:

```
# tar xvf /tmp/home.tar
```

To extract files from *files.tar* in the */tmp* directory:

```
# cd /tmp
# tar xvf /tmp/files.tar
```

To create a tarball called */tmp/home.tar.gz* of the */home* directory and compress it with *gzip*:

```
# tar cvzf /tmp/home.tar.gz /home
```

To create a tarball called */tmp/home.tar.bz2* of the */home* directory and compress it with *bzip2*:

```
# tar cvjf /tmp/home.tar.bz2 /home
```

To list both compressed archives and check which of the two is smaller:

```
# ll /tmp/home.tar*
-rw-r--r--. 1 root root  1481321  Nov 25 12:20  /tmp/home.tar.bz2
-rw-r--r--. 1 root root  1712239  Nov 25 12:20  /tmp/home.tar.gz
```

To create a tarball called */tmp/extattr.tar.bz2* of the files in */home* directory and include their extended attributes as well as SELinux contexts, and compress the archive with *bzip2*:

tar cvj --selinux --xattrs –f /tmp/extattr.tar.bz2 /home

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